Topic 7 – Rates of reaction and energy changes

Rates of reaction

Students should:		Maths skills
7.1	*Core Practical: Investigate the effects of changing the conditions of a reaction on the rates of chemical reactions by:* *a measuring the production of a gas (in the reaction between hydrochloric acid and marble chips)* *b observing a colour change (in the reaction between sodium thiosulfate and hydrochloric acid)*	1a, 1c 4a, 4b, 4c, 4d, 4e
7.2	Suggest practical methods for determining the rate of a given reaction	4b, 4c, 4d, 4e
7.3	Explain how reactions occur when particles collide and that rates of reaction are increased when the frequency and/or energy of collisions is increased	1c
7.4	Explain the effects on rates of reaction of changes in temperature, concentration, surface area to volume ratio of a solid and pressure (on reactions involving gases) in terms of frequency and/or energy of collisions between particles	1c, 1d 5c
7.5	Interpret graphs of mass, volume or concentration of reactant or product against time	1c 4a, 4d, 4e
7.6	Describe a catalyst as a substance that speeds up the rate of a reaction without altering the products of the reaction, being itself unchanged chemically and in mass at the end of the reaction	
7.7	Explain how the addition of a catalyst increases the rate of a reaction in terms of activation energy	
7.8	Recall that enzymes are biological catalysts and that enzymes are used in the production of alcoholic drinks	

Use of mathematics

- Arithmetic computation, ratio when measuring rates of reaction (1a and 1c).
- Drawing and interpreting appropriate graphs from data to determine rate of reaction (4b and 4c).
- Determining gradients of graphs as a measure of rate of change to determine rate (4d and **4e**).
- Proportionality when comparing factors affecting rate of reaction (1c).

Suggested practicals

- Investigate the effect of potential catalysts on the rate of decomposition of hydrogen peroxide.

Heat energy changes in chemical reactions

Students should:		Maths skills
7.9	Recall that changes in heat energy accompany the following changes: a salts dissolving in water b neutralisation reactions c displacement reactions d precipitation reactions and that, when these reactions take place in solution, temperature changes can be measured to reflect the heat changes	
7.10	Describe an exothermic change or reaction as one in which heat energy is given out	
7.11	Describe an endothermic change or reaction as one in which heat energy is taken in	
7.12	Recall that the breaking of bonds is endothermic and the making of bonds is exothermic	
7.13	Recall that the overall heat energy change for a reaction is: a exothermic if more heat energy is released in forming bonds in the products than is required in breaking bonds in the reactants b endothermic if less heat energy is released in forming bonds in the products than is required in breaking bonds in the reactants	
7.14	**Calculate the energy change in a reaction given the energies of bonds (in kJ mol^{-1})**	1a, 1c
7.15	Explain the term activation energy	
7.16	Draw and label reaction profiles for endothermic and exothermic reactions, identifying activation energy	4a

Use of mathematics

- Arithmetic computation when calculating energy changes (1a).
- Interpretation of charts and graphs when dealing with reaction profiles (4a).

Suggested practicals

- Measure temperature changes accompanying some of the following types of change:
  a salts dissolving in water
  b neutralisation reactions
  c displacement reactions
  d precipitation reactions.